A ladder of models for inverse bremsstrahlung absorption

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Abstract

The interatomic potential was built as a ladder: Coulomb, then Yukawa, which screens it, then Lennard-Jones, which adds a repulsive core, each rung keeping the one below as a limit and adding one physical length. Inverse bremsstrahlung (IB) absorption is organized the same way here, from the Drude model with the Spitzer logarithm through the quantum logarithm, screening by the electrons, the correlations of the ions, strong collisions and their cross term with the medium, the non-Maxwellian distribution, speckles, strong fields and the anisotropy; for neutral atoms through exact phase shifts, polarization bremsstrahlung and multiple collisions. The correlations of the ions are computed in the hypernetted-chain approximation for the ion mixture; for hydrogen they equal Debye–Hückel to 2×10^{-4}, and with them the medium lowers the absorption of the nitrogen–hydrogen mixtures by 2 to 7 %. The anisotropy of the heated distribution is computed with two logarithms and electron–electron collisions and lowers the absorption by 0.8 to 1.3 % for hydrogen. The heaters' own strong-field factor and the exact factor at $v_{\rm os} \geq v_t$ are computed. In argon the single-collision photon exchange beyond the formula of Kroll and Watson is computed exactly and equals double scattering in the target at 2 to 7×10^{13} cm^{-3}. Applied to the fourteen shots of Turnbull et al., the ladder, with no fitted parameter, gives 0.96 ± 0.04 of the measured absorption for the ten nitrogen–hydrogen shots and 1.19 ± 0.06 for the four hydrogen shots. The residual is confined to hydrogen, where every rung of the theory is small.

1 The idea

(1) Every IB model in use is the Drude absorption coefficient with a collision frequency [1],

$$K = \frac{\nu_{\rm eff}}{c}\,\frac{\omega_p^2}{\omega^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2}, \tag{1.1}$$

and the models differ only in $\nu_{\rm eff}$. For electrons on ions of charge Ze,

$$\nu_{\rm eff} = \nu_T\,\Lambda, \qquad \nu_T = \frac{4\sqrt{2\pi}}{3}\,\frac{n_i Z^2 e^4}{m^{1/2}(k_B T_e)^{3/2}}, \tag{1.2}$$

and everything sits in the dimensionless "logarithm" Λ and in factors multiplying it. For electrons on neutral atoms $\nu_{\rm eff}$ is built from the cross sections of the atom.
(2) Each rung below has four parts: the formula, the effect it adds, the dimensionless parameter that says when the effect matters, and its size at the reference conditions. Each rung reduces to the rung below when its parameter goes to zero. Gaussian units are used.
(3) *The plasma reference.* Turnbull et al. measured the absorption of a 527-nm probe beam in gas-jet plasmas heated by thirteen 351-nm beams, with the plasma conditions from Thomson scattering [2, 3]. Of their fifteen analysed shots, fourteen have published profiles: four in hydrogen and ten in a

mixture of 43 % nitrogen and 57 % hydrogen nuclei. All are read at the density peak of their profiles (Tables 1 and 2). The probe carried 1 to 4 J in 100 ps over a 165-μm spot, about 10^{14} W cm^{-2}, so that at the late shots the Langdon parameter is the probe's own; at the co-timed shots the probe carries the share w_p of the overlapped intensity. The Langdon parameter α is computed from the 351-nm-equivalent intensity plotted by Turnbull et al.; it agrees with the one that Matte's formula gives from their plotted exponent m to 5.2 % at every shot. Their composite, the exact (Sommerfeld) logarithm and the Langdon factor with the velocity-dependent Gaunt factor, without their screening factor, exceeds the measured absorption by a factor 1.29 ± 0.06 for the hydrogen shots and 1.07 ± 0.04 for the mixture shots, the averages and spreads they quote [3]. The *hydrogen point*, $n_e/n_c = 0.020$, $T_e = 0.36$ keV, $T_i = 0.05$ keV, serves for the rungs that do not depend on the field. *The gas reference* is argon, the main constituent of one fill of an electron-beam-pumped KrF laser, at electron temperatures of 300 K and 1.08 eV and at 1, 4 and 28.3 THz.

Table 1: The four hydrogen shots of Turnbull et al. [3] at the density peak: probe time, n_e/n_c at 527 nm, T_e, the Langdon parameter α, the probe's share w_p of the intensity, and $u = \hbar\omega/k_B T_e$. The ion temperature is 0.05 keV.

shot	t_{probe} (ns)	n_e/n_c	T_e (keV)	α	w_p	u
102727	0.6	0.0177	0.37	0.044	1	0.00636
104031	0.3	0.0229	0.34	0.119	0.27	0.00692
106310	0.3	0.0100	0.26	0.180	0.29	0.00905
106311	0.3	0.0047	0.20	0.232	0.29	0.01177

Table 2: The ten nitrogen–hydrogen shots of Turnbull et al. [3] at the density peak: probe time, n_e/n_c, T_e, T_i, the nitrogen charge Z_N used in their calculations, $Z_{\mathrm{eff}} = \langle Z^2\rangle/\langle Z\rangle$, the coupling Γ_{NN} of the nitrogen ions, α and w_p.

shot	t_{probe} (ns)	n_e/n_c	T_e (keV)	T_i (keV)	Z_N	Z_{eff}	Γ_{NN}	α (w_p)
102729	0.6	0.0149	0.65	0.070	6.5	5.57	0.276	0.086 (1.00)
102737	0.3	0.0179	0.42	0.050	5	4.16	0.261	0.189 (0.40)
102738	0.3	0.0189	0.70	0.070	6	5.10	0.260	0.547 (0.15)
104035	0.3	0.0179	0.62	0.070	5	4.16	0.186	0.299 (0.23)
104037	0.6	0.0179	0.62	0.100	6	5.10	0.179	0.067 (1.00)
104041	0.6	0.0194	0.43	0.090	6	5.10	0.204	0.108 (1.00)
106315	0.6	0.0184	0.63	0.090	6	5.10	0.201	0.147 (1.00)
106316	0.6	0.0080	0.45	0.045	5.5	4.63	0.261	0.150 (1.00)
106317	0.6	0.0039	0.30	0.030	5	4.16	0.261	0.196 (1.00)
106319	0.6	0.0279	0.80	0.120	7	6.04	0.225	0.086 (1.00)

2 The electron–ion ladder

2.1 Drude–Spitzer

The classical logarithm $\Lambda = \ln(b_{\max}/b_{\min})$ with the Debye length $b_{\max} = \lambda_{De} = v_t/\omega_p$ and $b_{\min}$ the larger of the 90° impact parameter $Ze^2/3k_BT_e$ and the de Broglie length $\hbar/2mv$, both at $mv^2 = 3k_BT_e$ [1]; $v_t = (k_BT_e/m)^{1/2}$. This is the "Coulomb potential" of the ladder: right in form, with cutoffs put in by hand. At the hydrogen point $\lambda_{De} = 15.73$ nm, the de Broglie length $0.2887\,\hbar/mv_t = 4.20 \times 10^{-3}$ nm exceeds $Ze^2/3k_BT_e = 1.33 \times 10^{-3}$ nm, and $\Lambda_0 = 8.23$.

2.2 The quantum logarithm

Second order in the ion potential, with Volkov (field-dressed) electron states and a Maxwell distribution, gives the collision frequency of Eq. (1.2) with

$$\Lambda_1(u) = \frac{1 - e^{-u}}{u}\, e^{u/2} K_0\Big(\frac{u}{2}\Big), \qquad u = \frac{\hbar\omega}{k_BT_e}, \tag{2.1}$$

absorption net of stimulated emission, and no cutoff. This is Kramers' result with the Born Gaunt factor, $\Lambda_1 = (\pi/\sqrt{3})\langle g_B\rangle(1 - e^{-u})/u$ with $\langle g_B\rangle = (\sqrt{3}/\pi)e^{u/2}K_0(u/2)$ [4, 5]. For $u \ll 1$, $K_0(x) = -\ln(x/2) - \gamma + O(x^2 \ln x)$ gives

$$\Lambda_1 \simeq \ln\frac{4k_BT_e}{e^\gamma\hbar\omega} = \ln\frac{b_{\max}}{b_{\min}}, \qquad b_{\max} = \frac{v_t}{\omega}, \qquad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t} = 0.445\,\frac{\hbar}{mv_t}, \tag{2.2}$$

with γ Euler's constant. The theory fixes both cutoffs. The upper one is the distance an electron travels in a radian of the laser period, the dynamic cutoff of Dawson and Oberman [6], not the Debye length. This is the "Yukawa" step: the long range is cut by a physical length. Parameter: ω/ω_p for the upper cutoff, u for the photon energy. At the hydrogen point $b_{\max} = 2.225$ nm and $\Lambda_1 = 5.84$. The logarithm Λ_0 of Sec. 2.1 exceeds it by $\ln(\omega/\omega_p) + \ln(0.445/0.2887) = 1.96 + 0.43$, or 40.9 %, in line with the finding of Turnbull et al. that transport logarithms with a Debye length miss their data by more than 50 % [2].

2.3 Dynamic screening by the electrons

With the electrons' own response, second order in the ion potential gives the high-frequency collision frequency of Dawson and Oberman [6],

$$\nu_{\text{eff}} = -\frac{2Ze^2}{3\pi m\omega}\int_0^\infty k^2 S(k)\,\mathrm{Im}\,\frac{1}{\varepsilon(k,\omega)}\,\mathrm{d}k, \tag{2.3}$$

with ε the longitudinal dielectric function of the electrons and S the static structure factor of the ions. For bare ions ($S = 1$) and the quantum (Lindhard) dielectric function of Maxwell electrons, with $x = kv_t/\omega$ and $r = \omega_p/\omega$,

$$\Lambda_2 = -\frac{1}{r^2}\Big(\frac{2}{\pi}\Big)^{1/2}\int_0^\infty x^2\,\mathrm{Im}\,\frac{1}{\varepsilon(x)}\,\mathrm{d}x, \qquad \varepsilon(x) = 1 - \frac{r^2}{\sqrt{2}\,ux^3}\big[Z(\zeta_+) - Z(\zeta_-)\big], \quad \zeta_\pm = \frac{1}{\sqrt{2}}\Big(\frac{1}{x} \pm \frac{ux}{2}\Big), \tag{2.4}$$

where Z is the plasma dispersion function. Replacing $\mathrm{Im}\,\varepsilon^{-1}$ by $-\mathrm{Im}\,\varepsilon$ gives back Eq. (2.1). Parameter: $r = \omega_p/\omega$. At the hydrogen point the electrons lower the logarithm by 0.0076, or 0.13 %.

2.4 Correlations of the ions

The ions enter Eq. (2.3) only through $S(k)$. Their mean spacing does not appear by itself; uncorrelated ions add whatever their spacing. In a weakly coupled plasma S has the Debye–Hückel form

$$S(k) = \frac{k^2 + k_{De}^2}{k^2 + k_{De}^2 + k_{Di}^2}, \qquad k_{Di}^2 = a\,k_{De}^2, \qquad a = \frac{Z T_e}{T_i}. \tag{2.5}$$

With static (Debye) electrons and soft photons the ion term has a closed form. The one-speed logarithm changes by $\int_{\omega/v}^{\infty}[S(q)-1]\,\mathrm{d}q/q = -(k_{Di}^2/2k_t^2)\ln(1+k_t^2v^2/\omega^2)$, $k_t^2 = k_{De}^2 + k_{Di}^2$, and the Maxwell average, with $\int_0^\infty e^{-s}\ln(1+\kappa s)\,\mathrm{d}s = e^{1/\kappa}E_1(1/\kappa)$, gives

$$\Delta\Lambda_3 \simeq \bar{I} = -\frac{a}{2(1+a)}\,e^{1/\kappa}E_1(1/\kappa), \qquad \kappa = 2(1+a)\,\frac{n_e}{n_c}. \tag{2.6}$$

Parameters: $k_{Di}/k_{De} = a^{1/2}$ for the size, and the ion coupling $\Gamma_{ii} = Z^2e^2/a_ik_BT_i$ for the validity of Eq. (2.5), with $a_i = (3/4\pi n_i)^{1/3}$. At the hydrogen point $a = 7.2$, $\kappa = 0.328$ and $\Gamma_{ii} = 0.020$. Equation (2.3) with Eq. (2.5) and the Lindhard ε lowers the logarithm by a further 0.114, or 1.96 %; Eq. (2.6) gives −0.1136, within 0.44 % of it. With the electron screening of Sec. 2.3, the medium lowers Λ_1 by 2.08 %. At the four shots the medium factor $\Lambda_{2,3}/\Lambda_1$ is 0.981, 0.978, 0.991 and 0.996 (Table 6); it would be 0.968 at the first shot with $T_i = 0.025$ keV and 0.989 with 0.1 keV.

Against the fitted factor. Turnbull et al. multiply the absorption by $f_{sc} = \{\ln[(2k_BT_e/\hbar\omega)\,y/(1+y^2)^{1/2}] - 1/[2(1+y^2)]\}/\ln(2k_BT_e/\hbar\omega)$, $y = l_{sc}\omega/v_t$, the Born logarithm of a Yukawa ion of screening length l_{sc} [2]. With $l_{sc} = \lambda_{D,ei} = 5.49$ nm, the Debye length of electrons and ions, it lowers the absorption by 2.56 %, close to the medium of Secs. 2.3 and 2.4. With the interionic distance $l_{sc} = a_i = 1.437$ nm, chosen to match their data, it lowers it by 16.8 %, about eight times what the medium gives.

Beyond Debye–Hückel. Equation (2.5) is the weak-coupling limit. The structure factor of an ion mixture follows at any coupling from the Ornstein–Zernike relation with the hypernetted-chain closure [26]. With ions i of charge Z_ie and density n_i at the temperature T_i, interacting through the electron-screened potential $\phi_{ij} = Z_iZ_je^2e^{-k_{De}r}/r$,

$$h_{ij}(r) = \exp\Big[-\frac{\phi_{ij}(r)}{k_BT_i} + h_{ij}(r) - c_{ij}(r)\Big] - 1, \qquad \tilde{h}_{ij}(k) = \tilde{c}_{ij}(k) + \sum_l \tilde{c}_{il}(k)\,n_l\,\tilde{h}_{lj}(k), \tag{2.7}$$

and the absorption takes the charge structure factor

$$S_{ZZ}(k) = \frac{\sum_i n_iZ_i^2 + \sum_{ij} n_in_jZ_iZ_j\tilde{h}_{ij}(k)}{\sum_i n_iZ_i^2} \tag{2.8}$$

in place of S in Eq. (2.3). The equations are solved by iteration on a radial grid, with the long-range parts of c_{ij} and $h_{ij}-c_{ij}$ subtracted and transformed analytically, so that the Debye–Hückel solution is the exact starting point. For the one-component plasma the excess energy is −0.5704 at $\Gamma = 1$, against −0.5726 from the Monte Carlo fit of Slattery, Doolen and DeWitt [27], and −0.02569 at $\Gamma = 0.1$, against −0.02590 from Abe's expansion [28] and −0.02739 from Debye–Hückel. At the hydrogen point the hypernetted-chain S differs from Eq. (2.5) by at most 6.2×10^{-4}, and at the four hydrogen shots the medium factor changes by at most 1.9×10^{-4}: for hydrogen the correlations beyond Debye–Hückel are nil. The mixtures are another matter. There $k_{Di}^2/k_{De}^2 = Z_{\mathrm{eff}}T_e/T_i$ is 24 to 52 and $\Gamma_{NN} = 0.18$ to 0.28, the medium factor is 0.930 to 0.982, and the hypernetted chain raises it by 0.0037 to 0.0064 against Debye–Hückel. The medium lowers the absorption by 2 to 7 %

in the mixtures and by 0.4 to 2.2 % in hydrogen. Halving or doubling T_i, the least certain reading, moves the mixture medium factor between 0.904 and 0.989. The fitted factor of Turnbull et al., with the interionic distance of the nitrogen ions for the mixtures as they use it and of the protons for hydrogen, is 0.879 to 0.969 for the mixtures and 0.827 to 0.927 for hydrogen; at the density peaks it reproduces their reported changes $1.29 \to 1.13$ and $1.07 \to 1.00$ to 0.8 and 2.3 %.

2.5 Strong collisions

The Born approximation needs $\eta = Ze^2/\hbar v \ll 1$. The exact Coulomb (Sommerfeld) free–free Gaunt factor removes that condition [7, 5]:

$$\Lambda_4 = \frac{\pi}{\sqrt{3}}\,\langle g\rangle(\gamma^2, u)\,\frac{1 - e^{-u}}{u}, \qquad \gamma^2 = \frac{Z^2\,\mathrm{Ry}}{k_B T_e} = \frac{\eta_t^2}{2}, \quad \eta_t = \frac{Ze^2}{\hbar v_t}, \tag{2.9}$$

with the thermally averaged exact Gaunt factor $\langle g\rangle$ tabulated by van Hoof et al. [8]. As $\gamma^2 \to 0$ it tends to $\langle g_B\rangle$ and Λ_4 to Λ_1. In the soft-photon limit the exact logarithm of one electron is the Born one plus Bloch's constant $\Delta(\eta) = -\gamma - \mathrm{Re}\,\psi(1 + i\eta)$ [9], which is $-\zeta(3)\eta^2$ for $\eta \ll 1$ and $-\gamma - \ln\eta$ for $\eta \gg 1$, where it turns the quantum lower cutoff into the classical one. Parameter: η_t. At the hydrogen point $\eta_t = 0.275$ and $\langle g\rangle/\langle g_B\rangle = 0.99315$, a correction of -0.685 %; at the four shots it is -0.67 to -0.88 %. For fully stripped nitrogen ($Z = 7$) at the same T_e and u, $\eta_t = 1.92$ and the correction is -15.4 %.

The cross term with the medium. The exact correction is added to the logarithm of Sec. 2.4, $\Lambda_{2,3} + (\Lambda_4 - \Lambda_1)$. Whether the medium changes the strong-collision correction itself is decided by the partial waves of an ion screened to $V = -(Ze^2/r)e^{-r/\lambda}$. In the soft-photon limit the logarithm of one electron is the transport sum $S = \sum_l (l+1)\sin^2(\delta_{l+1} - \delta_l)$ of its phase shifts, cut at the partial wave $J = kb_{\max} = k^2/\omega$ of the dynamic cutoff, in units of η^2; with the first Born phases, $S^B = \sum_l (l+1)(\delta^B_{l+1} - \delta^B_l)^2$. For the bare Coulomb field $\delta_l = \mathrm{ph}\,\Gamma(l + 1 - i\eta)$, $\delta^B_l = -\eta\psi(l+1)$, and $(S - S^B)/\eta^2 = \Delta(\eta) + O(\eta^2/J^2)$ exactly. For the screened field the exact phases come from the radial equation, integrated numerically, and the Born phases are $\delta^B_l = \eta\,Q_l(1 + 1/2k^2\lambda^2)$ with the Legendre function of the second kind. The cross term

$$C = \frac{S^Y - S^{Y,B}}{\eta^2} - \frac{S^C - S^{C,B}}{\eta^2} \tag{2.10}$$

is the part of the exact-minus-Born correction that the screening changes; the Born part of the screening, $(S^{Y,B} - S^{C,B})/\eta^2$, is the second-order medium of Sec. 2.4 and is not in C. For the Coulomb field only odd powers of η enter the phases, so there is no second-Born term in their differences; for the screened field there is, and C is first order in $b_{\max}/\lambda$, not in b_{90}/λ as the size of the strong-collision region would suggest. Averaged over the Maxwellian with the soft-photon weight e^{-s}, $s = E/k_B T_e$, at $T_e = 0.36$ keV: for hydrogen $\langle C\rangle = +0.0098$ with $\lambda = \lambda_{D,ei} = 5.49$ nm and $+0.0035$ with $\lambda = \lambda_{De} = 15.73$ nm, against $\langle\Delta\rangle = -0.134$ for the bare ion, that is 7.4 % and 2.6 % of the strong-collision correction and 1.7×10^{-3} and 6.0×10^{-4} of Λ_1. For $Z = 7$, $\langle C\rangle = +0.0359$ and $+0.0128$ against $\langle\Delta\rangle = -1.235$, that is 2.9 % and 1.0 % of the correction and 6.1×10^{-3} and 2.2×10^{-3} of Λ_1. The sign is positive: the medium weakens the strong-collision correction. Doubling the cutoff J changes C by 9 % at $s = 0.25$ and by 2 to 3 % at $s \geq 1.2$. The soft-photon average $\langle\Delta\rangle$ overstates the exact thermal correction of Eq. (2.9) by a factor 3.3 for hydrogen, since the slow electrons, for which the photon is not soft, are focused by the ion; C shares that weighting, so these are upper values. In the absorption the cross term is below 0.2 % for hydrogen and about 0.6 % for the nitrogen ions of the mixed shots. At the conditions of the mixture shots, $Z = 6$, $T_e = 0.62$ keV and the total screening length $\lambda_t = 3.58$ nm of shot 106315, $\langle C\rangle = +0.0383$ against $\langle\Delta\rangle = -0.870$, 6.0×10^{-3} of Λ_1. This closes the item.

2.6 The non-Maxwellian distribution (Langdon)

IB heats slow electrons fastest. When it outpaces the electron–electron collisions the distribution flattens toward $F \propto \exp[-(v/v_m)^M]$ with $M > 2$ [10]. The parameter is $\alpha = Zv_{\rm os}^2/v_t^2$, with $v_{\rm os}$ the peak quiver velocity, and Matte et al. fit the exponent as $m(\alpha) = 2 + 3/(1 + 1.66/\alpha^{0.724})$ [11]. For an isotropic distribution the heating rate is $P \propto -\int_0^\infty L(v)\,\partial_v F\,{\rm d}v$, with the one-speed Born logarithm $L(v) = \ln(2mv^2/\hbar\omega)$. At fixed L this is $LF(0)$, and the factor is $F_M(0)/F_2(0)$ at fixed density and energy; Langdon's fit $f_L = 1 - 0.553/[1 + (0.27/\alpha)^{0.75}]$ [10] equals it at $M = m(\alpha)$ to within 0.27 % for $\alpha = 10^{-3}$ to 10^3. With the velocity-dependent L, the substitution $s = (v/v_m)^M$ and $\int_0^\infty e^{-s}\ln s\,{\rm d}s = -\gamma$ give

$$\frac{P_M}{P_2} = \frac{F_M(0)}{F_2(0)}\,\frac{\ln\left(2v_m^2/uv_t^2\right) - 2\gamma/M}{\ln(4/u) - \gamma}, \qquad \frac{v_m^2}{v_t^2} = \frac{3\Gamma(3/M)}{\Gamma(5/M)}. \tag{2.11}$$

The second factor, one at $M = 2$, raises the absorption: a flattened distribution moves electrons to speeds where the logarithm is larger. At the four shots $m = 2.18$, 2.34, 2.45 and 2.52, $f_L = 0.887$, 0.806, 0.765 and 0.739, the logarithm factor C_L is 1.031, 1.056, 1.072 and 1.085, and the product $f_L C_L$ is 0.914, 0.851, 0.820 and 0.802. At the mixture shots $\alpha = 0.067$ to 0.547.

2.7 Speckles

A real beam has an intensity pattern. If the distribution adjusts locally and T_e is uniform, the absorbed power is that of the mean intensity times an average of f_L over the pattern. The probe's own pattern has the exponential distribution of a fully developed speckle field, $x = I/\langle I\rangle \sim e^{-x}$, and the probe is absorbed where its own intensity is, with the weight x; the thirteen heater beams carry independent patterns, whose sum y has the gamma distribution of shape 13 and mean one. With $\alpha_p = w_p\alpha$ and $\alpha_h = (1 - w_p)\alpha$,

$$f_{sp} = \int_0^\infty {\rm d}x\, x\, e^{-x} \int_0^\infty {\rm d}y\, \frac{13^{13} y^{12} e^{-13y}}{12!}\, f_L(\alpha_h y + \alpha_p x). \tag{2.12}$$

Since f_L falls with α, the bright speckles absorb less per unit intensity. Parameter: α, and the ratio of the speckle size to the electron relaxation length (Eq. (2.12) is the local limit; in the opposite limit $f_{sp} = f_L$). At the four shots $f_{sp}/f_L = 0.955$, 0.977, 0.973 and 0.972: the late shot, where the probe is the only beam, is the most affected.

2.8 Strong fields: the probe's own field and the heaters' quiver

When the quiver velocity is not small against the thermal velocity the field cannot be treated to lowest order. Second order in the ion potential but exact in the field gives the heating rate as a sum over photon numbers ℓ, each weighted by $J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)$ with $\mathbf{a}_0$ the quiver amplitude, inside the dielectric form of Secs. 2.3 and 2.4 [12, 13, 14]. Parameter: $\beta = v_{\rm os}/v_t = (\alpha/Z)^{1/2}$. At the four shots the probe has $\beta_p = 0.21$, 0.18, 0.23 and 0.26, and the heaters $\beta_h = 0$, 0.29, 0.36 and 0.40. Two effects follow, computed here with the free-electron response χ_0 of Maxwell electrons and bare ions, in the variables $x = kv_t/\omega_p$, $u = \hbar\omega_p/k_BT_e$ and

$$D(x;\nu) = e^{-\frac{1}{2}(\nu/x - ux/2)^2} - e^{-\frac{1}{2}(\nu/x + ux/2)^2}, \qquad \int_0^\infty D(x;1)\,\frac{{\rm d}x}{x} = u\Lambda_1(u), \tag{2.13}$$

D being proportional to $x\,{\rm Im}\,\chi_0(k, \nu\omega_p)$, absorption net of emission.

The probe alone. With $B_\ell(z)=\int_{-1}^{1}J_\ell^2(z\mu)\,\mathrm{d}\mu$ the exact ratio of the probe's collision frequency to its weak-field value is

$$R_{\mathrm{self}}(\beta_p,u)=\frac{3}{u\beta_p^2\Lambda_1}\sum_{\ell=-\infty}^{\infty}\ell\int_0^\infty\frac{\mathrm{d}x}{x^3}\,D(x;\ell)\,B_\ell(x\beta_p),\tag{2.14}$$

which tends to one as $\beta_p\to0$. Its quasi-static limit at a fixed logarithm is Silin's $1-\frac{9}{40}\beta_p^2+O(\beta_p^4)$; the exact coefficient at $u=0.0064$ is 0.195, smaller because the largest momentum transfer grows with the field. At the four shots $R_{\mathrm{self}}=0.991$, 0.994, 0.990 and 0.987.

The probe inside the heaters' quiver. With two fields the quiver is $\mathbf{a}(t)=\mathbf{a}_h\sin\omega_h t+\mathbf{a}_p\sin\omega_p t$, and the Volkov phase $e^{i\mathbf{k}\cdot\mathbf{a}(t)}$ has sidebands at $\ell\omega_h+m\omega_p$ with the weights $J_\ell^2(\mathbf{k}\cdot\mathbf{a}_h)J_m^2(\mathbf{k}\cdot\mathbf{a}_p)$; for $\omega_h/\omega_p=3/2$ no two sidebands with $|m|\le1$ coincide. The probe's photons are counted by m, and to second order in the probe field only $m=\pm1$ enter, so that

$$R_{pp}(\psi)=\frac{\displaystyle\sum_{\ell=-\infty}^{\infty}\int_0^\infty\frac{\mathrm{d}x}{x}\,D\Big(x;1+\ell\frac{\omega_h}{\omega_p}\Big)\,W_\ell^{\psi}\Big(\frac{x\beta_h\omega_p}{\omega_h}\Big)}{\dfrac{2}{3}\displaystyle\int_0^\infty\frac{\mathrm{d}x}{x}\,D(x;1)},\tag{2.15}$$

$$W_\ell^{\psi}(z)=\int_{-1}^{1}J_\ell^2(z\mu)\Big[\frac{\sin^2\psi}{2}(1-\mu^2)+\cos^2\psi\,\mu^2\Big]\mathrm{d}\mu,\tag{2.16}$$

with ψ the angle between the two polarizations; averaged over ψ, $W_\ell=B_\ell/3$, and $W_\ell^{\psi}(0)=\frac{2}{3}\delta_{\ell0}$ returns the weak field. The terms with $\ell<0$ have $1+\ell\omega_h/\omega_p<0$ and $D<0$: an electron that absorbs a probe photon while emitting a heater photon loses energy, so the reverse process, a heater photon absorbed and a probe photon emitted, is favoured, and the heater feeds the probe through the ions. At large k the sidebands cancel, since $\sum_\ell J_\ell^2=1$ and $\sum_\ell\ell J_\ell^2=0$ and D is linear in ν there; the effect is a logarithmic integral from the thermal cutoff $x\sim1$ upward, of relative size β_h^2. Two independent checks: the small-field series of Eq. (2.15), from $B_0=2-z^2/3$ and $B_{\pm1}=z^2/6$, gives $R_{pp}=1-c\beta_h^2$ with $c=0.212$ averaged over ψ at $u=0.0064$, which the full sum reproduces to 6×10^{-5} at $\beta_h=0.2$; and the classical picture of ions drifting through the electron gas with the velocity $-(\mathbf{u}_h+\mathbf{u}_p)(t)$, a drag $-h(s)\mathbf{s}$ with $s=u/\sqrt{2}v_t$ and $h=[\sqrt{\pi}\,\mathrm{erf}(s)/s-2e^{-s^2}]/s^2$, gives for the probe's power $\langle F\cdot u_p\rangle$ to second order $R_{pp}^{qs}(\psi)=\frac{3}{4}\langle h(s_h)+s_hh'(s_h)\cos^2\psi\rangle$, which at small β_h is $1-\frac{9}{20}\beta_h^2$ for parallel polarizations, $1-\frac{3}{20}\beta_h^2$ for perpendicular ones and $1-\frac{1}{4}\beta_h^2$ averaged. A pump is therefore twice as effective on a probe as a beam on itself, $\frac{9}{40}\to\frac{9}{20}$, because the two phases are independent. At $\beta_h=0.2$ the exact $1-R_{pp}$ is 0.0151, 0.0051 and 0.0084 for the three cases against 0.0178, 0.0060 and 0.0099 quasi-static, the same ratio 0.847 in all three, as for R_{self} against Silin. The thirteen heaters surround the plasma, so the average over ψ is the physical case. At the co-timed shots $R_{pp}=0.982$, 0.974 and 0.967 averaged, 0.968, 0.953 and 0.941 for parallel and 0.989, 0.984 and 0.980 for perpendicular polarizations. The product $R_{\mathrm{self}}R_{pp}$ is 0.991, 0.976, 0.964 and 0.954 at the four shots. For the heater beams themselves the single-beam factor at their own $\beta_h=0.29$, 0.36 and 0.40 is $R_{\mathrm{self}}=0.983$, 0.976 and 0.969 at the co-timed shots: the heating rate is 2 to 3 % below its weak-field value. At $\beta\ge1$ Eq. (2.14) at $u=0.0064$ gives 0.953, 0.828 and 0.507 for $\beta=0.5$, 1 and 2, against Silin's 0.946, 0.805 and 0.460: the exact rate exceeds the quasi-static one by 1, 3 and 10 %, because the largest momentum transfer grows with the quiver velocity. This closes the item.

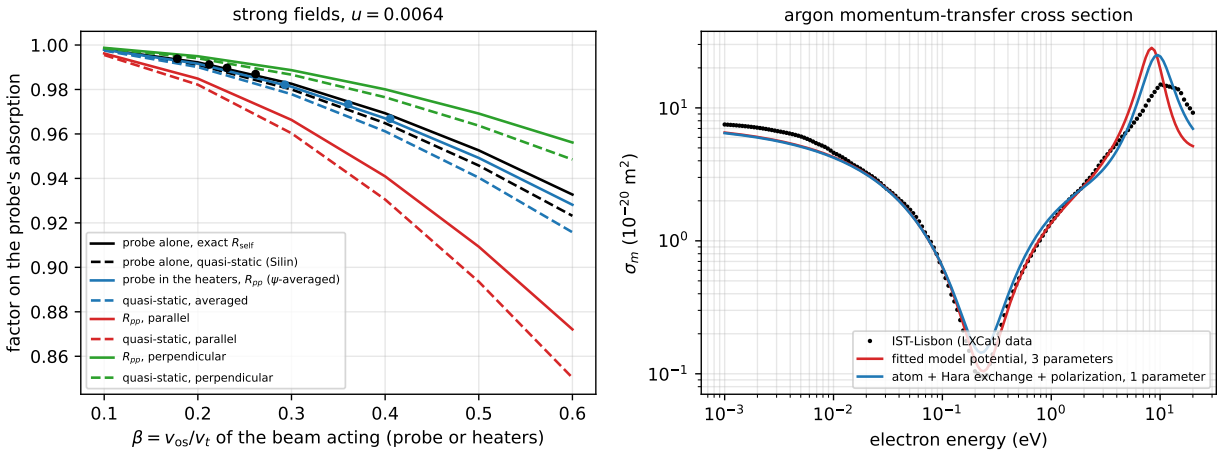


Figure 1: Left: the strong-field factors on a probe's absorption at $u = 0.0064$ against $\beta = v_{os}/v_t$ of the beam that acts. Black: the probe's own field, exact (Eq. (2.14)) and quasi-static (Silin). Colours: the probe inside the heaters' quiver, Eq. (2.15), averaged over the polarizations (blue), parallel (red) and perpendicular (green), exact (solid) and in the quasi-static drag picture (dashed). Dots: the four hydrogen shots. Right: the momentum-transfer cross section of argon from the IST-Lisbon data, from the three-parameter model potential of Eq. (3.2), and from the Hartree–Fock atom with Hara's exchange and one fitted cutoff.

2.9 The anisotropy

A linearly polarized field heats along its polarization, and the distribution acquires a part of degree 2 in the angle, $f = f_0 + f_2(v)P_2(\hat{\mathbf{v}}\cdot\boldsymbol{\epsilon})$. Its direct effect on the absorption of the same beam is

$$\frac{\delta P}{P} = -\frac{2}{5}\,\frac{v_{os}^2}{v_t^2}\,Q, \qquad Q = \int_0^\infty \frac{f_2(v)}{(v_{os}^2/3v_t^2)\,f_0(0)}\,\frac{\mathrm{d}v}{v}, \tag{2.17}$$

with v_{os} the probe's own quiver velocity. In the Lorentz limit, $Z \gg 1$, the field drives f_2 through the ion collisions and the same collisions relax it, $f_2 = S_2/3\nu_{ei}$, and for the self-similar heated distribution Q is a number of order one, 1.32, 1.03, 0.93 and 0.87 at the α of the four shots. Two corrections enter. The drive is inverse bremsstrahlung and carries the logarithm Λ_1 with the dynamic cutoff, while the relaxation of the quasi-static f_2 is pitch-angle scattering and carries the transport logarithm $\ln(\lambda_{De}/b_{\min})$ of Sec. 2.1, 8.10 to 8.36 at the shots: f_2 is smaller by $\Lambda_1/\Lambda_{tr} = 0.63$ to 0.71. And at $Z = 1$ the electron–electron collisions relax f_2 as well. With the test-particle collision operator on a Maxwellian background [29], in units of $4\pi n_e e^4\Lambda_{tr}/m^2v_t^3$ and with $w = v/v_t$, $x = w^2/2$ and $\psi(x) = \mathrm{erf}\sqrt{x} - 2(x/\pi)^{1/2}e^{-x}$, f_2 solves

$$3\Big(\frac{Z}{w^3} + \nu_D\Big)f_2 - \frac{1}{w^2}\frac{\mathrm{d}}{\mathrm{d}w}\Big[w^2\frac{\psi}{w^3}\Big(\frac{\mathrm{d}f_2}{\mathrm{d}w} + wf_2\Big)\Big] = \frac{3Z}{w^3}\,\frac{\Lambda_1}{\Lambda_{tr}}\,S_2, \qquad \nu_D = \frac{(1-1/2x)\psi + \psi'}{w^3}, \tag{2.18}$$

with the drive S_2 of the Lorentz limit. The operator keeps the Maxwellian, and for fast electrons at $Z = 1$ the deflection by electrons equals that by ions, as it must [30]. For a Maxwellian drive the electron–electron collisions lower Q by a factor 0.77 to 0.80 at the hydrogen shots and 0.94 to 0.96 at the mixture shots, where $Z_{\mathrm{eff}} \approx 5$. Applied to the self-similar Q, the direct effect is 1.30, 0.76, 1.02 and 1.15 % at the four hydrogen shots, against 2.3, 1.3, 2.0 and 2.4 % in the Lorentz limit with one logarithm, and 0.32 to 1.07 % at the mixture shots. The feedback of f_2 on the isotropic part and the field-particle part of the electron–electron operator are not computed; the direct part is.

2.10 The plasma ladder at the hydrogen point

Table 3: The electron–ion ladder at the hydrogen point ($n_e/n_c = 0.020$, $T_e = 0.36$ keV, $T_i = 0.05$ keV, 526.7 nm). The logarithm after each rung and its change against Λ_1.

Sec.	Adds	Parameter	Λ	against Λ_1 (%)
2.1	Drude–Spitzer, cutoffs by hand	—	8.228	+40.9
2.2	quantum logarithm, dynamic cutoff	u, ω/ω_p	5.839	0
2.3	screening by the electrons	ω_p/ω	5.832	−0.13
2.4	correlations of the ions	$a^{1/2}$, Γ_{ii}	5.717	−2.08
2.5	strong collisions	η_t	5.678	−2.77
2.5	cross term with the medium	$\eta_t\, b_{\max}/\lambda$	5.681 to 5.687	−2.71 to −2.60

The rungs of Secs. 2.6 to 2.9 multiply the absorption rather than the logarithm (Sec. 4).

3 The electron–atom ladder

In a weakly ionized gas the electrons absorb mostly while they collide with atoms. For ω much larger than the momentum-transfer collision frequency $\nu_m(v) = n_a \sigma_m(v) v$ the Lorentz-gas conductivity gives, for any isotropic F normalized to one, $\nu_{\text{eff}} = -(4\pi/3)\int_0^\infty v^3 \nu_m(v)\, \partial_v F \, \mathrm{d}v$.

3.1 Drude with a mean collision frequency

$\nu_{\text{eff}} = \langle \nu_m \rangle = n_a \langle \sigma_m v \rangle$ over the Maxwellian. This is the usual input of a Drude model.

3.2 The energy-dependent kernel

For a Maxwellian, $\partial_v F = -(v/v_t^2)F$ and

$$\nu_{\text{eff}} = \frac{\langle \nu_m v^2 \rangle}{3 v_t^2} = \left\langle \nu_m \, \frac{2\varepsilon}{3 k_B T_e} \right\rangle, \tag{3.1}$$

with $\varepsilon = m v^2/2$, which reduces to $\langle \nu_m \rangle$ of Sec. 3.1 only for a constant ν_m. The ratio to $\langle \nu_m \rangle$ is 4/3 for a constant σ_m. The IST-Lisbon momentum-transfer cross section of argon [15, 16, 17] has its Ramsauer minimum near 0.25 eV, and with it the ratio is 0.836 at 300 K and 2.02 at 1.08 eV. A Drude model with a mean collision frequency is wrong by up to a factor of two here. Parameter: the logarithmic slope of $\sigma_m(\varepsilon)$. By the low-energy theorem of Low [18] and the formula of Kroll and Watson [19], this kernel is exact for soft photons, $\hbar\omega \ll \varepsilon$, and the atom enters only through σ_m.

3.3 Exact phase shifts

When $\hbar\omega$ is comparable with the electron energy the free–free cross section depends on more than σ_m. It is computed from the acceleration-form dipole matrix elements between exact partial waves

of a model potential, here

$$V(r) = -\frac{Z}{r}\,\frac{1}{1+H(e^{r/d}-1)} - \frac{\alpha_d}{2(r^2+r_c^2)^2} \tag{3.2}$$

in atomic units, with the screening function of Green, Sellin and Zachor [20], $Z = 18$ and the polarizability $\alpha_d = 11.083$ [21]. Its parameters, $H = 4.321$, $d = 0.7869$ and $r_c = 0.8266$, minimize the rms of $\ln\sigma_m$ against the IST-Lisbon set over 36 energies from 0.003 to 4 eV (rms 0.081), and it binds the three s and two p orbitals of argon. With $G(\varepsilon)$ the ratio of the exact cross section to its soft-photon limit, the ratio of the quantum to the classical absorption coefficient is

$$R = \frac{1-e^{-u}}{u}\,\frac{\langle\sigma_m G\rangle_w}{\langle\sigma_m\rangle_w}, \tag{3.3}$$

with the weight $\varepsilon^2 e^{-\varepsilon/k_BT_e}$ of Eq. (3.1). Parameter: u. For argon $R = 0.9984$ at 1 THz and 0.9749 at 4 THz for 300 K electrons, and 0.591 at 28.3 THz, the CO_2 laser; at 1.08 eV it stays within 10^{-3} of one at all three frequencies.

From the atom. Equation (3.2) is fitted to σ_m with three parameters, so the mid-infrared value rests on the fit. A potential built from the atom replaces it: the static field of the closed-shell Hartree–Fock atom, $V_{st} = -Z/r + V_H$, whose total energy is -526.831 against the Hartree–Fock limit -526.818 hartree, plus a local exchange, plus the same polarization tail with the single parameter r_c. The semiclassical exchange of Furness and McCarthy, $V_{ex} = \frac{1}{2}(E-V_{st}) - \frac{1}{2}[(E-V_{st})^2 + 4\pi\rho]^{1/2}$ with ρ the electron density [22], is far too attractive at these energies: no r_c brings its σ_m within a factor of 4.5 of the data. The free-electron-gas exchange of Hara [23],

$$V_{ex} = -\frac{k_F}{\pi}\Big[1 + \frac{1-\eta^2}{2\eta}\ln\Big|\frac{1+\eta}{1-\eta}\Big|\Big], \qquad k_F = (3\pi^2\rho)^{1/3}, \quad \eta = \frac{(k^2+2I+k_F^2)^{1/2}}{k_F}, \tag{3.4}$$

with I the ionization energy, does reproduce the Ramsauer region: with r_c =1.548 the rms of $\ln\sigma_m$ over the same 36 energies is 0.172, against 0.081 for the three-parameter fit, and the minimum lies at 0.226 eV against about 0.25 eV in the data; over 0.003 to 20 eV the rms is 0.215. With this potential R at 300 K is 0.9986 at 1 THz, 0.9744 at 4 THz and 0.566 at 28.3 THz, against 0.9984, 0.9749 and 0.591 for Eq. (3.2); at 1.08 eV it is 1.0009 at 28.3 THz. The exchange depends on the energy, and the matrix elements between k and k' use the potential at the mean energy. The mid-infrared reduction is therefore a property of the phase shifts that any potential reproducing σ_m shares, not of the fit.

3.4 Polarization bremsstrahlung

The field of the electron induces a dipole $\alpha_d\hat{\mathbf{r}}/r^2$ (atomic units) in the atom, which radiates together with the electron. In the acceleration form, where the electron's dipole matrix element is that of $V'(r)/\omega^2$, the radial force $V'(r)$ becomes $V'(r) - \omega^2\alpha_d/(r^2+r_c^2)$, with the cutoff r_c of the polarization tail. Parameter: $\alpha_d\omega^2$ against the atomic force. For argon with Eq. (3.2) it changes R by $-0.03\,\%$ at 1 THz and $-0.43\,\%$ at 4 THz for 300 K electrons, and by $-13.0\,\%$ at 28.3 THz, where R becomes 0.514; at 1.08 eV the change is below $0.1\,\%$. With the potential from the atom, R at 28.3 THz and 300 K becomes 0.491.

3.5 Multiple collisions

An electron can scatter on two atoms in a row. The kinetic equation counts every collision with its own photon-exchange cross section whatever collision follows, so sequences of collisions add nothing

to the absorption of a gas. They do add a signal proportional to the square of the density in laser-assisted scattering experiments, which may explain why photon exchange in argon in CO_2-laser fields was measured far above the Kroll–Watson prediction [24, 25]. That anomaly is a question about single-collision cross sections, and measurements at several target densities would settle it.

Table 4: The electron–atom ladder for argon. $\nu_{\rm eff}/\langle\nu_m\rangle$ is the ratio of the collision frequency of Eq. (3.1) to that of Sec. 3.1; R is Eq. (3.3) without and with polarization bremsstrahlung (Secs. 3.3 and 3.4), for the fitted potential of Eq. (3.2) and for the potential from the atom with Hara's exchange.

			fitted potential		atom + Hara exchange	
T_e	$\nu_{\rm eff}/\langle\nu_m\rangle$	f (THz)	R (Sec. 3.3)	R (Sec. 3.4)	R (Sec. 3.3)	R (Sec. 3.4)
300 K	0.836	1	0.9984	0.9981	0.9986	0.9983
		4	0.9749	0.9707	0.9744	0.9702
		28.3	0.591	0.514	0.566	0.491
1.08 eV	2.02	1	1.0000	1.0000	1.0000	1.0000
		4	1.0000	1.0000	1.0000	1.0001
		28.3	1.0008	1.0018	1.0009	1.0019

3.6 One collision against two

The anomaly of Sec. 3.5 needs the size of the single-collision photon exchange where Kroll and Watson give zero, $\boldsymbol{\epsilon}\cdot\mathbf{Q}=0$. To first order in the field and exactly in the atomic potential, the one-photon amplitude between the scattering states of momenta $\mathbf{k}_i$ and $\mathbf{k}_f$ is, in the acceleration form and atomic units,

$$M=\sum_{l,\,l'=l\pm1} i^{\,l-l'}e^{i(\delta^f_{l'}+\delta^i_l)}R_{l'l}\sum_{mm'}Y_{l'm'}(\hat{\mathbf{k}}_f)\,\langle l'm'|\boldsymbol{\epsilon}\cdot\hat{\mathbf{r}}|lm\rangle\,Y^*_{lm}(\hat{\mathbf{k}}_i),\qquad R_{l'l}=\frac{1}{\omega^2}\int_0^\infty u_{k_f l'}V' u_{k_i l}\,\mathrm{d}r, \tag{3.5}$$

with radial waves of unit amplitude, and the ratio of the one-photon signal to the elastic one in a field of amplitude E_0 is

$$\frac{\mathrm{d}\sigma_{+1}}{\mathrm{d}\sigma_{\rm el}}=\frac{16\pi^2E_0^2|M|^2}{k_i^3k_f\,|f_{\rm el}|^2}, \tag{3.6}$$

with $f_{\rm el}$ the elastic amplitude. Three properties check it. Averaged over the polarization and integrated over $\hat{\mathbf{k}}_f$, $|M|^2$ gives back the integrated free–free cross section of Sec. 3.3, to 5×10^{-14}. At 0.3 THz, where the photon is soft, Eq. (3.6) with $\boldsymbol{\epsilon}\parallel\mathbf{Q}$ equals the Kroll–Watson value $(k_f/k_i)(\boldsymbol{\epsilon}\cdot\mathbf{Q})^2a_0^2/4$ to 6.4×10^{-4}. In the geometry with $\boldsymbol{\epsilon}$ normal to the scattering plane it vanishes to 10^{-31}, as the reflection symmetry of Sec. 3.5 requires. The partial waves are taken to $l=22$, with the polarization tail of the elastic amplitude summed in the Born approximation, $\delta_l=\pi\alpha_d k^2/[(2l-1)(2l+1)(2l+3)]$, beyond [31]; going to $l=26$ changes the signal by 4.8×10^{-3} and halving the radial step by 3.5×10^{-5}. In the expansion in the field the next term carries $a_0k\,\hbar\omega/E\approx0.03$ at 10 eV.

In the in-plane geometry, $\boldsymbol{\epsilon}$ in the scattering plane and perpendicular to $\mathbf{Q}$, argon at 10 eV in a CO_2 laser at 10^8 W cm^{-2} gives a one-photon signal of 2.25×10^{-3} of the elastic one at 90° with

the fitted potential of Eq. (3.2) and 1.23×10^{-3} with the potential from the atom; 1.72×10^{-3} and 1.00×10^{-3} at 8 eV, 3.80×10^{-4} and 3.37×10^{-4} at 20 eV. This is the dipole amplitude f_D of the in-plane geometry, 37 and 20 times the nondipole term, which with the laser beam along $\mathbf{Q}$ gives $(k_f/k_i)J_1^2(x_1) = 6.09 \times 10^{-5}$ with $x_1 = a_0(\hat{\mathbf{k}} \cdot \mathbf{Q})(\boldsymbol{\epsilon} \cdot \bar{\mathbf{v}})/c = 0.0156$, $\bar{\mathbf{v}}$ the mean of the two velocities. To first order in v/c the in-plane amplitude is $f_D + f_N \cos\varphi$, with φ the angle between the laser direction $\hat{\mathbf{k}}$ and $\mathbf{Q}$ (the reflection through the scattering plane removes a term in $\sin\varphi$), so the interference $2\mathrm{Re}(f_D^* f_N)\cos\varphi$ can move the in-plane signal by up to $2|f_N/f_D|$, 33 and 44 % of $|f_D|^2$, as the laser direction turns. The polarization bremsstrahlung of Sec. 3.4 changes the in-plane signal by at most 0.6 %, since its first order vanishes with $\boldsymbol{\epsilon} \cdot \mathbf{Q}$. In absolute terms the single-collision cross section at 10 eV stays below 1.3 % of the elastic one at 90° at every angle; its ratio to the elastic signal rises only where the elastic cross section has a minimum, to 1.19×10^{-2} and 0.18 at 120°, and there the two potentials disagree.

Double scattering in the target is computed with the same potentials, for the gas cylinder of radius 1 mm and length 2 mm about the beam: each collision follows Kroll and Watson, the second collision returns the photon count to one, and the signal per incident electron is $n^2 D_1$ with

$$D_1 = \int \mathrm{d}\Omega_1 \, G(\hat{\mathbf{u}}_1) \, \frac{\mathrm{d}\sigma_{\mathrm{el}}}{\mathrm{d}\Omega}(\hat{\mathbf{z}} \cdot \hat{\mathbf{u}}_1) \, \frac{\mathrm{d}\sigma_{\mathrm{el}}}{\mathrm{d}\Omega}(\hat{\mathbf{u}}_1 \cdot \hat{\mathbf{u}}_d) \sum_\ell J_\ell^2(\xi) J_{1-\ell}^2(\xi), \qquad \xi = k a_0 \, \boldsymbol{\epsilon} \cdot (\hat{\mathbf{u}}_1 - \hat{\mathbf{z}}), \tag{3.7}$$

where G is the path length in the gas integrated along the beam. Relative to the single-collision elastic signal it is $n d_1$, with $d_1 = D_1/(L \, \mathrm{d}\sigma_{\mathrm{el}}/\mathrm{d}\Omega)$. At 90°, $d_1 = 3.27 \times 10^{-17}$ and 2.77×10^{-17} cm^3 in the in-plane geometry and 4.50×10^{-17} and 4.26×10^{-17} cm^3 in the normal geometry. The single collision and the double scattering are therefore equal at $n^* = 6.9 \times 10^{13}$ and 4.5×10^{13} cm^{-3} at 90°, at 5.3 and 2.8×10^{13} cm^{-3} at 60° and at 2.4 and 2.9×10^{13} cm^{-3} at 120°: at the densities of the targets used, the two mechanisms are of the same size in the in-plane geometry. In the normal geometry the single collision is absent. Both rise toward the minima of the elastic cross section, both stay one to two orders of magnitude below the measured 12 % away from them, and a density scan in the two geometries separates them: an intercept in the in-plane geometry only, a slope in both. The two potentials give σ_m(10 eV) 33 and 63 % above the data, so the numbers at 10 eV carry that uncertainty; their crossover densities n^* agree within a factor 2.

Table 5: Argon at 28.3 THz and 10^8 W cm^{-2}: the exact single-collision one-photon signal relative to the elastic one in the in-plane geometry with $\boldsymbol{\epsilon} \perp \mathbf{Q}$, double scattering d_1 in the same geometry, and the density n^* at which the two are equal, for the fitted potential and the potential from the atom, at 10 eV.

	fitted potential			atom + Hara exchange		
angle	single	d_1 (cm^3)	n^* (cm^{-3})	single	d_1 (cm^3)	n^* (cm^{-3})
60°	2.93×10^{-3}	5.51×10^{-17}	5.32×10^{13}	1.61×10^{-3}	5.79×10^{-17}	2.79×10^{13}
90°	2.25×10^{-3}	3.27×10^{-17}	6.89×10^{13}	1.23×10^{-3}	2.77×10^{-17}	4.45×10^{13}
120°	1.19×10^{-2}	5.05×10^{-16}	2.36×10^{13}	1.81×10^{-1}	6.36×10^{-15}	2.85×10^{13}

4 Putting the rungs together

4.1 The two kinds of scatterer add

Electron–ion and electron–atom collisions are separate binary events, and electron–electron collisions radiate no dipole radiation, since two electrons have the same charge-to-mass ratio. So $\nu_{\rm eff} = \nu_T\Lambda + \nu_{ea}$. In argon at 1 atm and 300 K, at 1 THz, the two are equal at $n_e = 1.0\times10^{13}$ cm^{-3} for 300 K electrons, the Ramsauer minimum making the atoms weak scatterers, and at 1.0×10^{16} cm^{-3} for 1.08 eV electrons, close to the critical density 1.24×10^{16} cm^{-3} at that frequency.

4.2 The standard model

Collecting the rungs, the absorption coefficient of a probe is Eq. (1.1) with

$$\boxed{\begin{aligned}\nu_{\rm eff} = \nu_T\Big[\Lambda_{2,3}(r,u,\{Z_i\}) + \Delta\Lambda_4(\eta_t,u) + C\Big]\, f_L(\alpha)\, C_L\big(m(\alpha),u\big) \\ \times\, \frac{f_{sp}}{f_L}\, R_{\rm self}(\beta_p,u)\, R_{pp}(\beta_h,u)\left(1+\frac{\delta P}{P}\right) \\ +\; n_a\Big\langle \sigma_m v\, \frac{2\varepsilon}{3k_BT_e}\Big\rangle R(u),\end{aligned}} \tag{4.1}$$

where $\Lambda_{2,3}$ is Eq. (2.3) in the form of Eq. (2.4) with the charge structure factor of the ion mixture from Eq. (2.7), $\Delta\Lambda_4 = \Lambda_4 - \Lambda_1$, C is the cross term of Eq. (2.10), C_L is the second factor of Eq. (2.11), f_{sp} is Eq. (2.12), $R_{\rm self}$ and R_{pp} are Eqs. (2.14) and (2.15), $\delta P/P$ is Eq. (2.17) with f_2 from Eq. (2.18), and the last term is Eq. (3.1) times R of Secs. 3.3 and 3.4. Products of factors are a first-order approximation in each correction.

4.3 The budgets of the fourteen shots

The ladder shares the rungs of Secs. 2.2, 2.5 and 2.6 with the composite of Turnbull et al. What it adds is the product

$$X = \frac{\Lambda_{2,3} + \Delta\Lambda_4 + C}{\Lambda_1 + \Delta\Lambda_4}\, \frac{f_{sp}}{f_L}\, R_{\rm self}\, R_{pp}\left(1+\frac{\delta P}{P}\right), \tag{4.2}$$

listed shot by shot in Tables 6 and 7. The exact-Gaunt correction $\Delta\Lambda_4$ and the cross term C are weighted by $n_iZ_i^2$ over the ion species: $\Delta\Lambda_4$ is -0.67 to -0.88 % of Λ_1 for hydrogen and -8.3 to -12.1 % for the mixtures, where the nitrogen ions carry it, and $C = +0.0098$ for the protons and $+0.0383$ for the nitrogen ions. For hydrogen X is 0.916 to 0.926, with a mean of 0.921; for the mixtures it is 0.871 to 0.911, with a mean of 0.895.

4.4 Rules against counting an effect twice

1. *One upper cutoff.* The dynamic cutoff v_t/ω of Sec. 2.2 and the Debye length of Sec. 2.1 are two models of the same long-range cut. Electron screening enters through ε in Sec. 2.3, never as a second cutoff.

2. *One account of the ions.* The rung of Sec. 2.4 and a fitted screening factor both describe how the ions modify the long-range part of the logarithm. Applying f_{sc} on top of that rung counts the ion correlations twice.

Table 6: The budget of the four hydrogen shots: the logarithm factor of Eq. (4.2), the speckle factor, the strong-field factors, the anisotropy, their product X of Eq. (4.2) and the ladder against the measurements, $1.29X$. The fitted factor f_{sc} of Turnbull et al. is listed for comparison.

shot	logarithm	f_{sp}/f_L	R_{self}	R_{pp}	$-\delta P/P$	X	ladder/data	f_{sc}
102727	0.983	0.955	0.991	1	0.0130	0.918	1.185	0.837
104031	0.979	0.977	0.994	0.982	0.0076	0.926	1.195	0.827
106310	0.993	0.973	0.990	0.974	0.0102	0.922	1.189	0.884
106311	0.998	0.972	0.987	0.967	0.0115	0.916	1.181	0.927

Table 7: The budget of the ten nitrogen–hydrogen shots, as Table 6; the ladder against the measurements is $1.07X$.

shot	logarithm	f_{sp}/f_L	R_{self}	R_{pp}	$-\delta P/P$	X	ladder/data	f_{sc}
102729	0.944	0.946	0.997	1	0.0049	0.886	0.948	0.911
102737	0.949	0.965	0.996	0.994	0.0047	0.903	0.966	0.915
102738	0.936	0.986	0.997	0.981	0.0032	0.899	0.962	0.896
104035	0.949	0.977	0.997	0.988	0.0037	0.910	0.973	0.898
104037	0.953	0.949	0.997	1	0.0045	0.898	0.961	0.904
104041	0.957	0.944	0.996	1	0.0064	0.894	0.956	0.917
106315	0.949	0.942	0.994	1	0.0080	0.881	0.943	0.902
106316	0.968	0.942	0.994	1	0.0084	0.898	0.961	0.943
106317	0.987	0.942	0.991	1	0.0107	0.911	0.975	0.969
106319	0.928	0.946	0.997	1	0.0047	0.871	0.932	0.879

3. *Strong collisions once.* The exact Gaunt factor of Sec. 2.5 already contains the Born term; it is added as $\Lambda_4 - \Lambda_1$ to the medium logarithm, not multiplied by it, and the cross term of Eq. (2.10) is the only further piece.

4. *One logarithm throughout.* The Langdon factor of Sec. 2.6 must use the same velocity-dependent logarithm as Sec. 2.2 or 2.5; Langdon's fit alone assumes a fixed one.

5. *The field once per beam.* R_{self} holds the probe's own quiver, R_{pp} the heaters'; the Langdon parameter α of Sec. 2.6 takes the overlapped intensity because it describes the time-averaged distribution, which every beam shapes. Neither replaces the other.

6. *No separate term for collision sequences.* Double scattering is already in the kinetic equation (Sec. 3.5).

7. *Electron–electron collisions shape, they do not absorb.* They enter through α in Sec. 2.6 and through the relaxation of f_2 in Sec. 2.9, not as a third scatterer.

8. *Two logarithms in the anisotropy.* The field drives f_2 with the inverse-bremsstrahlung logarithm, and collisions relax it with the transport logarithm; using one for both overstates the anisotropy by $\Lambda_{\mathrm{tr}}/\Lambda_1 \approx 1.5$.

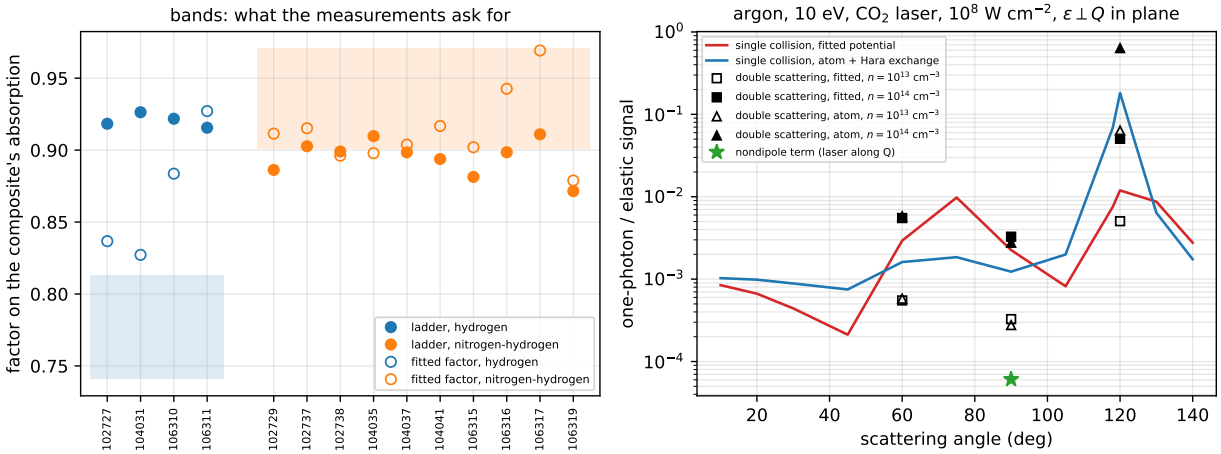


Figure 2: Left: per shot, the factor X of Eq. (4.2) that the ladder applies beyond the composite of Turnbull et al. (filled) and their fitted screening factor f_{sc} (open), against the factor the measurements ask for, $1/(1.29 \pm 0.06)$ for hydrogen and $1/(1.07 \pm 0.04)$ for the mixtures (bands). Right: argon at 10 eV in a CO_2 laser at 10^8 W cm^{-2}, in the in-plane geometry with $\boldsymbol{\epsilon} \perp \mathbf{Q}$: the exact single-collision one-photon signal relative to the elastic one for the two potentials (lines), double scattering in the target at 10^{13} and 10^{14} cm^{-3} (markers), and the nondipole term (star).

5 Have the models improved?

Plasma. Against the composite of Turnbull et al. without their screening factor, which is 1.29 ± 0.06 times the measured absorption of the hydrogen shots and 1.07 ± 0.04 times that of the mixture shots, the ladder is

$$\frac{K_{\text{ladder}}}{K_{\text{measured}}} = \begin{cases} 1.07\,\bar{X}_{\text{N}_2/\text{H}_2} = 1.07 \times 0.895 = 0.96 \pm 0.04 & \text{(ten nitrogen–hydrogen shots)}, \\ 1.29\,\bar{X}_{\text{H}_2} = 1.29 \times 0.921 = 1.19 \pm 0.06 & \text{(four hydrogen shots)}. \end{cases} \tag{5.1}$$

For the mixtures the ladder, with no fitted parameter, agrees with the measurements as well as the fitted composite of Turnbull et al., 1.00 ± 0.04 [3]. In these plasmas the fitted factor and the computed correlations of the ions do the same thing, 0.879 to 0.969 against a medium factor of 0.930 to 0.982, because the heavy ions make the correlations strong. For hydrogen they part: the fitted factor lowers the absorption by 7 to 17 %, the medium by 0.4 to 2.2 %, and the hypernetted chain confirms Debye–Hückel to 2×10^{-4} at $\Gamma_{ii} = 0.02$. *The ladder explains the mixture shots and leaves the hydrogen shots 19 % too absorbing.* Every rung that could act on hydrogen alone is now computed and small: the anisotropy, which scales as $1/Z$, 0.8 to 1.3 %; the strong fields 1 to 5 %; the speckles 2.3 to 4.5 %. The residual is the size of a 12 % error in T_e or a 9 % error in n_e of the hydrogen shots, whose measured absorption, 0.15 to 1.48 %, is the smallest of the data set (0.47 to 5.05 % for the mixtures).

The discriminating measurement. Helium against hydrogen at the conditions of the four hydrogen shots, each normalized by the unscreened model of its own charge: the ladder predicts 0.987, 0.985, 0.993 and 0.997, the fitted factor 1.042, 1.044, 1.034 and 1.024. The two differ by 2.7 to 5.8 % and move in opposite directions.

Gas. Against a Drude model with a mean collision frequency,

$$\boxed{\frac{K_{\text{ladder}}}{K_{\text{Drude}}} = \frac{\langle \nu_m \, 2\varepsilon/3k_B T_e \rangle}{\langle \nu_m \rangle} \, R = 0.836 \text{ (300 K) to } 2.02 \text{ (1.08 eV) at 1 to 4 THz,}} \tag{5.2}$$

where R of Secs. 3.3 and 3.4 is 0.97 to 1.00 at 1 to 4 THz, and at 28.3 THz R lowers it further, to 0.514 of the classical kernel for 300 K electrons with the fitted potential and 0.491 with the potential from the atom (Table 4). The single collisions beyond Kroll and Watson are computed (Table 5); they equal double scattering at the densities of the targets used.

6 What remains open

1. *The hydrogen residual.* The ladder accounts for the mixture shots and leaves 19 % for hydrogen, with every hydrogen-specific rung computed. The theory therefore places the residual in the hydrogen measurements or their plasma conditions, not in the absorption physics; the helium measurement of Sec. 5 decides between this and a fitted screening length.

2. *The anisotropy beyond its direct part.* The feedback of f_2 on the isotropic distribution and the field-particle part of the electron–electron operator are not computed. They change the 0.8 to 1.3 % of Sec. 2.9 by a factor of order one, not the residual.

3. *Argon at the elastic minima.* The single-collision and double-scattering signals are computed; near the minima of the elastic cross section, where the measured photon exchange is largest, both depend on the depth and the position of the minimum, which the two potentials do not agree on. A density scan in the in-plane and normal geometries measures both.

Closed here: the ion correlations beyond Debye–Hückel (Sec. 2.4), the strong-collision cross term at the mixtures' conditions (Sec. 2.5), the heaters' own strong-field factor and the exact factor at $\beta \geq 1$ (Sec. 2.8), the anisotropy with two logarithms and electron–electron collisions (Sec. 2.9), and the single-collision photon exchange beyond Kroll and Watson in argon (Sec. 3.6).

References

[1] S. Atzeni and J. Meyer-ter-Vehn, *The Physics of Inertial Fusion* (Oxford University Press, Oxford, 2004), doi:10.1093/acprof:oso/9780198562641.001.0001.

[2] D. Turnbull *et al.*, Phys. Rev. Lett. **130**, 145103 (2023), doi:10.1103/PhysRevLett.130.145103.

[3] D. Turnbull *et al.*, Phys. Plasmas **31**, 063304 (2024), doi:10.1063/5.0203446.

[4] H. A. Kramers, Phil. Mag. **46**, 836 (1923), doi:10.1080/14786442308565244.

[5] W. J. Karzas and R. Latter, Astrophys. J. Suppl. **6**, 167 (1961), doi:10.1086/190063.

[6] J. Dawson and C. Oberman, Phys. Fluids **5**, 517 (1962), doi:10.1063/1.1706652.

[7] A. Sommerfeld, Ann. Phys. **403**, 257 (1931), doi:10.1002/andp.19314030302.

[8] P. A. M. van Hoof, R. J. R. Williams, K. Volk, M. Chatzikos, G. J. Ferland, M. Lykins, R. L. Porter and Y. Wang, Mon. Not. R. Astron. Soc. **444**, 420 (2014), doi:10.1093/mnras/stu1438.

[9] F. Bloch, Ann. Phys. (Leipzig) **408**, 285 (1933), doi:10.1002/andp.19334080303.

[10] A. B. Langdon, Phys. Rev. Lett. **44**, 575 (1980), doi:10.1103/PhysRevLett.44.575.

[11] J. P. Matte, M. Lamoureux, C. Möller, R. Y. Yin, J. Delettrez, J. Virmont and T. W. Johnston, Plasma Phys. Control. Fusion **30**, 1665 (1988), doi:10.1088/0741-3335/30/12/004.

[12] H.-J. Kull and L. Plagne, Phys. Plasmas **8**, 5244 (2001), doi:10.1063/1.1416182.

[13] Th. Bornath, M. Schlanges, P. Hilse and D. Kremp, Phys. Rev. E **64**, 026414 (2001), doi:10.1103/PhysRevE.64.026414.

[14] C. D. Decker, W. B. Mori, J. M. Dawson and T. Katsouleas, Phys. Plasmas **1**, 4043 (1994), doi:10.1063/1.870874.

[15] L. L. Alves, J. Phys.: Conf. Ser. **565**, 012007 (2014), doi:10.1088/1742-6596/565/1/012007; IST-Lisbon database, `https://www.lxcat.net`.

[16] C. Yamabe, S. J. Buckman and A. V. Phelps, Phys. Rev. A **27**, 1345 (1983), doi:10.1103/PhysRevA.27.1345.

[17] L. C. Pitchford *et al.*, Plasma Process. Polym. **14**, 1600098 (2016), doi:10.1002/ppap.201600098.

[18] F. E. Low, Phys. Rev. **110**, 974 (1958), doi:10.1103/PhysRev.110.974.

[19] N. M. Kroll and K. M. Watson, Phys. Rev. A **8**, 804 (1973), doi:10.1103/PhysRevA.8.804.

[20] A. E. S. Green, D. L. Sellin and A. S. Zachor, Phys. Rev. **184**, 1 (1969), doi:10.1103/PhysRev.184.1.

[21] P. Schwerdtfeger and J. K. Nagle, Mol. Phys. **117**, 1200 (2019), doi:10.1080/00268976.2018.1535143.

[22] J. B. Furness and I. E. McCarthy, J. Phys. B: At. Mol. Phys. **6**, 2280 (1973), doi:10.1088/0022-3700/6/11/021.

[23] S. Hara, J. Phys. Soc. Jpn. **22**, 710 (1967), doi:10.1143/JPSJ.22.710.

[24] B. Wallbank and J. K. Holmes, J. Phys. B: At. Mol. Opt. Phys. **27**, 5405 (1994), doi:10.1088/0953-4075/27/21/027.

[25] I. Rabadán, L. Méndez and A. S. Dickinson, J. Phys. B: At. Mol. Opt. Phys. **29**, L801 (1996), doi:10.1088/0953-4075/29/22/004.

[26] T. Morita, Prog. Theor. Phys. **20**, 920 (1958), doi:10.1143/PTP.20.920.

[27] W. L. Slattery, G. D. Doolen and H. E. DeWitt, Phys. Rev. A **26**, 2255 (1982), doi:10.1103/PhysRevA.26.2255.

[28] R. Abe, Prog. Theor. Phys. **22**, 213 (1959), doi:10.1143/PTP.22.213.

[29] M. N. Rosenbluth, W. M. MacDonald and D. L. Judd, Phys. Rev. **107**, 1 (1957), doi:10.1103/PhysRev.107.1.

[30] L. Spitzer and R. Härm, Phys. Rev. **89**, 977 (1953), doi:10.1103/PhysRev.89.977.

[31] T. F. O'Malley, L. Spruch and L. Rosenberg, J. Math. Phys. **2**, 491 (1961), doi:10.1063/1.1703735.